Naga Venkata Sahithi Parepalli

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Summary

Computer Science and Engineering student with hands-on experience in Java development, Object-Oriented Programming, Data Structures and Algorithms, and SQL, with knowledge of backend development and REST APIs. Additional experience in AI/ML and Quantum Computing through practical projects. Led a team to secure 1st place in the Amaravati Quantum Valley Hackathon semifinals and qualify for the Grand Finale.

Skills

Languages: Java, Python, SQL, JavaScript
Software Development: OOP, Data Structures & Algorithms, DBMS, REST APIs, Backend Development
AI/ML: Machine Learning, Deep Learning, NLP, Generative AI, Computer Vision, RAG
Frameworks & Libraries: Spring Boot, FastAPI, Flask, TensorFlow, PyTorch, Scikit-learn, OpenCV
Tools & Platforms: Git, GitHub, Docker, MLflow, Google AI Studio, Vertex AI, IBM watsonx.ai
Quantum Computing: Qiskit, PennyLane, QAOA

Projects

TrustLens.AI – Multimodal Deepfake Detection Platform
Python, FastAPI, Computer Vision, OpenCV, MediaPipe, Signal Processing

- Developed a **multimodal media forensics platform** using **Python and FastAPI** to analyze manipulated images, videos, and audio.
- Implemented **FFT, Grad-CAM**, facial artifact analysis, and audio signal features with detailed reporting for **explainable media verification**.

AI-PriceOptima – Intelligent Dynamic Pricing System
Python, LightGBM, XGBoost, Machine Learning

- Developed a **machine learning-based dynamic pricing solution** using **LightGBM and XGBoost** to predict and optimize product prices.
- Evaluated predictive models on historical pricing data to support **data-driven pricing decisions**.

Hybrid Quantum Portfolio Optimizer
Python, Qiskit, QAOA, PennyLane, CVXPY, PySide6

- Led team **Quantum Coders** in building a **hybrid quantum-classical portfolio optimizer** using **QAOA** with a classical mean-variance comparison.
- Qualified for the **AQVH Grand Finale** and secured **first place** in the semi-final round.

Internship

Java Developer **Present**
ElevateIQ Soft Tech

- Completed **Java development** assignments focused on practical software development tasks.
- Applied **Object-Oriented Programming** and **exception handling** to write structured, maintainable programs.
- Practiced **debugging** and **problem-solving** to strengthen core Java skills.

Achievements

- **AQVH Quantum Computing Challenge:** Secured **1st place** in the semifinals, qualified for the Grand Finale, and received an **INR 10,000** prize.
- **WISER Program Selection (2025):** Selected among the top **3,000** of **65,000+** applicants for the WISER (BQPhy®) Quantum Computing Program.
- **WISER Summer Program 2026:** Selected for the fully funded **$1,000** program, with participants from 68 countries.
- **Quantum Innovation Excellence Award (2026):** Awarded at QC^2, a 36-hour national-level Quantum Computing Hackathon at SRM University AP (with GeeksforGeeks Student Chapter), for innovation in quantum solutions.
- **Infosys Springboard Pragati Cohort 5:** Selected for a 10-week industry-oriented internship program.

Education

PSCMR College of Engineering and Technology **2023 – 2027**
B.Tech in Computer Science and Engineering, CGPA: 8.9/10.0 *Vijayawada, India*

Intermediate (MPC) **2021 – 2023**
Mathematics, Physics and Chemistry, Percentage: 91% *Andhra Pradesh, India*

Class X **2020 – 2021**
Secondary School Education, Percentage: 98% *Andhra Pradesh, India*